Stovetop Monitor
Overview and Theory of Operation

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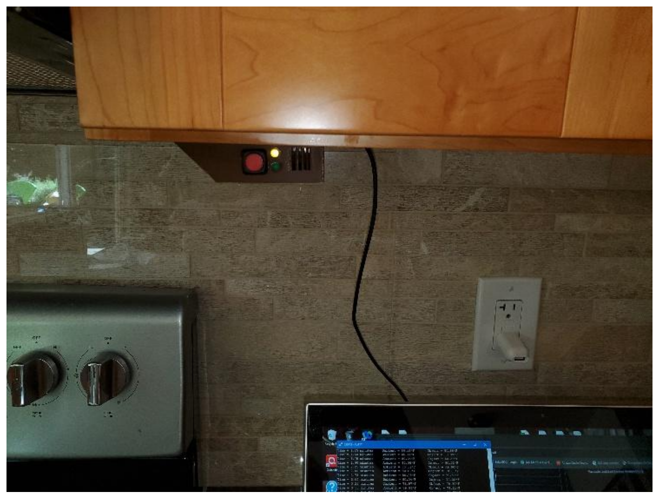
https://github.com/TeamPracticalProjects/Stovetop_Monitor/tree/main/Documents/Terms_of_Use_License_and_Disclaimer.pdf



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DOCUMENT OVERVIEW.

This document provides an overview of the Stovetop Monitor project from Team Practical Projects. The Stovetop Monitor monitors the state of a stovetop and alarms if it senses that a burner is turned on for too long.

PROBLEM STATEMENT AND SOLUTION OVERVIEW.

A stovetop burner that is accidentally left on presents a fire hazard. Busy/distracted people, as well as elderly people who are becoming forgetful, can accidentally leave a stovetop burner on, unattended. In the worst case, this can lead to a fire hazard and burn down the house as well as risking lives. In less than the worst case, it can lead to damage to the stove and/or cookware.

There are few, if any, stovetop/ovens on the market that provide remote access via an app. In general, interconnected stoves have apps that can monitor and control an oven but do not have a similar capability vis-a-vis the stovetop burners. These apps are aimed at remote control of baking in the oven and it is assumed that cooking on the stovetop will always be attended.

Modern stovetop/ovens tend not to have mechanical controls. Rather, they are electronically controlled via an embedded microcontroller. This means that the status of the stovetop burners cannot be monitored nor can they be controlled remotely by hacking into electro-mechanical switches. An external solution - one that does not modify the stovetop/oven in any way - needs to be developed.

The Stovetop Monitor project provides standoff, non-invasive monitoring of a stovetop with algorithms that determine if one or more burners are left on unintentionally. An infrared (IR) temperature sensor is used to detect, categorize and track the IR radiation from the stovetop. The difference between the IR temperature measurement from the stovetop area and the ambient temperature is used to classify the current operation of the stovetop into:

- WARMing
- COOKing
- BURNing

Each of these stovetop states is pre-assigned a time limit above which it can be assumed to exceed normal stovetop operation. When the time limit is exceeded in any of these states, an ALARM is indicated and the user is notified to take action to attend to the situation. The temperature (temperature difference) and timeout thresholds that ultimately trigger alarms are contained in a separate file so that installation-specific values can be used. The Stovetop

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Monitor allows a computer running a serial monitor to attach and record IR and ambient temperature profiles for various test conditions in order to calibrate the system for the specific type of stovetop and sensor position for any specific installation.

The specific calibration values released with this project were empirically determined for an electric glasstop stove and a sensor that is mounted under a cabinet that is above and off to the right side of the stovetop. The sensor has a 90 degree field of view (FOV) and looks down at the stovetop at a 45 degree angle. The one sensor therefore covers the state of all of the burners on a stovetop. In addition to electric stovetops, this project should work well with gas burner stovetops. This project is not expected to be useful for induction stovetops, which turn off automatically when a pan is removed from a burner.

The standoff and non-invasive nature of this problem solution is, necessarily, imperfect. We have opted to err on the side of false positives, as the consequences of too many alarms is annoying but the consequences of missed alarm conditions can be very severe.

STOVETOP MONITOR SYSTEM OVERVIEW.

Figure 1 presents a system diagram for the Stovetop Monitor project.

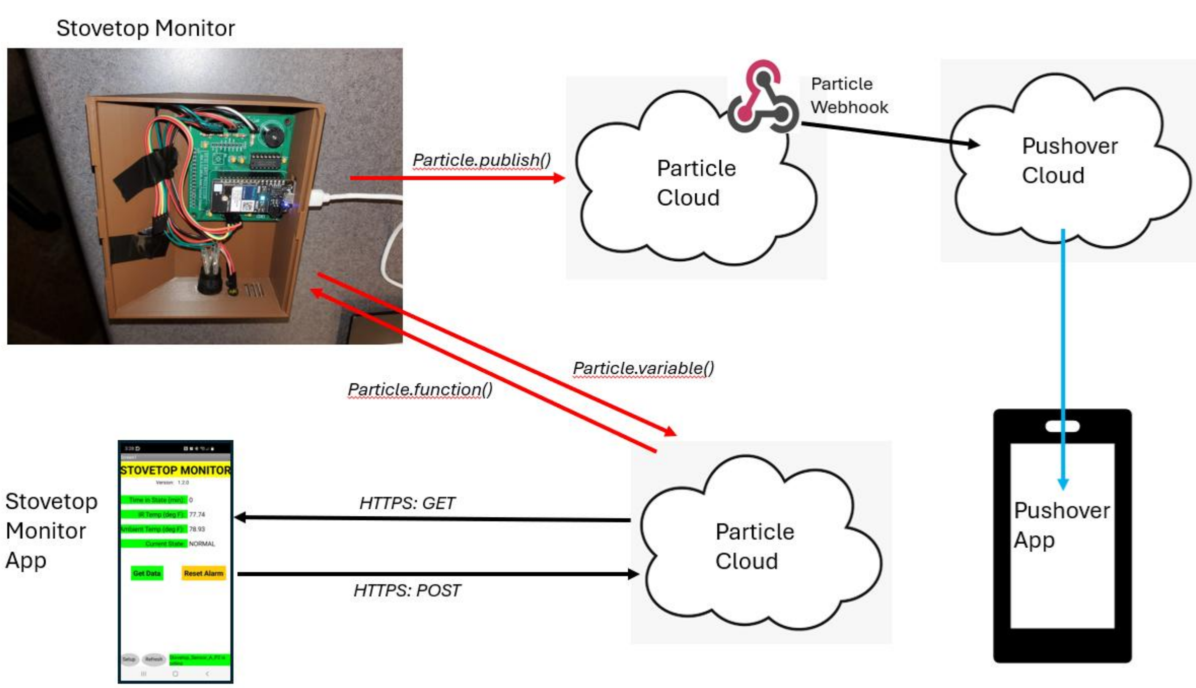

Figure 1. Stovetop Monitor System Overview.

The Stovetop Monitor electronics uses an IR temperature sensor to measure the infrared energy given off from the stovetop, as well as the ambient temperature at the sensor. A WiFi

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enabled microcontroller[1] uses these measurements to determine the state of the stovetop and to track the time in the current state. If the time in the current state exceeds a pre-determined value, an alarm is generated.

Alarm events are published to the Particle Cloud. A webhook in the Particle Cloud subscribes to alarm events and uses the Pushover API to generate a push notification on the user's smartphone.

An Android app uses the Particle Cloud API to read out the data from the electronics so that the user can determine the status of the Stovetop Monitor at any time. The App also allows the user to remotely reset any alarm that is in process.

Electronics.

The Stovetop Monitor electronics is based upon a Melexis MLX-60914 IR temperature sensor; see:

https://www.melexis.com/en/documents/documentation/datasheets/datasheet-mlx90614

We chose to use a breadboard friendly module that incorporates this sensor:

https://www.amazon.com/dp/B0B63K5V7T?ref=ppx_yo2ov_dt_b_fed_asin_title

The module includes the sensor itself, a low dropout voltage regulator, and I2C pullup resistors. The low dropout voltage regulator allows the module to operate with both 5 volt and 3.3 volt systems.

A Particle[2] Photon 2 microcontroller module forms the "brains" of the electronics. The Photon 2 is WiFi enabled and tightly integrated with the Particle Cloud. The Particle Cloud provides Internet connectivity to the electronics.

The sensor connects to the Photon 2 via the latter's I2C bus. The electronics also contains level shifter circuitry that allows the Photon 2 to control a green, yellow and red LED and a piezo buzzer, and connects a pushbutton switch as a digital input to the Photon 2. The electronic components are mounted on the printed circuit board that was developed as part of an earlier Team practical Projects project, see:

https://github.com/TeamPracticalProjects/MN_ACL/tree/master/Hardware/PC_board

Only part of the capability of this printed circuit board is used on this project. See the "Build and Install Manual" for this project for details.

[1] Particle Photon 2

[2] Particle.io

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The complete electronics are housed in a custom 3D printed enclosure with an accompanying 3D printed mounting bracket:

https://github.com/TeamPracticalProjects/Stovetop_Monitor/tree/main/Hardware

Particle Cloud.

The Particle Cloud provides Internet connectivity for this project:

- The firmware[3] running on the Photon 2 publishes alarm events to the Particle Cloud. A custom webhook[4] installed in the user's Particle account subscribes to these event publications and uses the Pushover[5] API to generate a push notification on the user's smartphone. The Pushover notification serves as a remote alarm notification that is available to the user anywhere that they have cell phone connectivity.

- The firmware running on the Photon 2 exposes several global variables to the Particle Cloud, including the firmware version, the current stovetop state, the latest IR temperature reading, the latest ambient temperature reading, and the accumulated time in the current stovetop state.

- The firmware running on the Photon 2 provides remote calling of a function in the firmware that resets any alarm that is in process.

Pushover.

Pushover is a commercial service for adding push notifications to any smartphone. The user must install the Pushover app on their smartphone (available for both iOS and Android). The app requires a one time in-app fee of $5 and supports up to 10,000 push notifications per month at no additional cost. The Pushover app allows the user to select from a range of priority values and a large set of notification sounds.

[3] https://github.com/TeamPracticalProjects/Stovetop_Monitor/tree/main/Software/StovetopMonitor

[4] https://github.com/TeamPracticalProjects/Stovetop_Monitor/tree/main/Software/Webhook

[5] https://pushover.net/

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Stovetop Monitor App.

A Stovetop Monitor App[6] is provided with this project. This App allows the user to monitor the status of the Stovetop Monitor and to remotely reset any alarm that is in process. The App is written in MIT App Inventor 2 has been built and tested for Android[7] devices. The intention is that the user can respond to a remote alarm by viewing the status of the system in order to gain a better understanding of the nature of the alarm and the urgency of a response. However, the non-invasive nature of this project does not provide a means for the user to remotely control the stovetop itself. If a burner has been left on unintentionally, it will need to be manually turned off at the stovetop itself.

The user can, alternatively, use the Particle Console to monitor the stovetop and reset alarms. See the “Stovetop Monitor User Manual” for further details.

STOVETOP MONITOR THEORY OF OPERATION.

The firmware running on the Photon 2 microcontroller reads data from the Melexis sensor once every 15 seconds. This time interval was selected because of the large thermal lag of the stovetop. More frequent readings just add noise to the algorithm, while meaningful temperature changes occur more slowly.

Each reading from the sensor updates:

- The current IR temperature measurement.
- The current ambient sensor temperature measurement.

These readings are used to update the state of the system, per the state diagram shown in figure 2, below.

Note that the Melexis sensor occasionally and randomly returns an erroneous reading of 1899.59 degrees F on either the IR temperature or the ambient temperature. The reason for these erroneous readings is not known, but we think it is due to the fact that the Melexis sensor uses a special subset of the I2C protocol that powers down automatically and needs to occasionally wake-up when queried for new data. The firmware detects and disregards these erroneous readings.

[6] https://github.com/TeamPracticalProjects/Stovetop_Monitor/tree/main/Software/AI2_App

[7] In principle, this app can be built for iOS and deployed via the Apple App Store; However, we have not tested this deployment.

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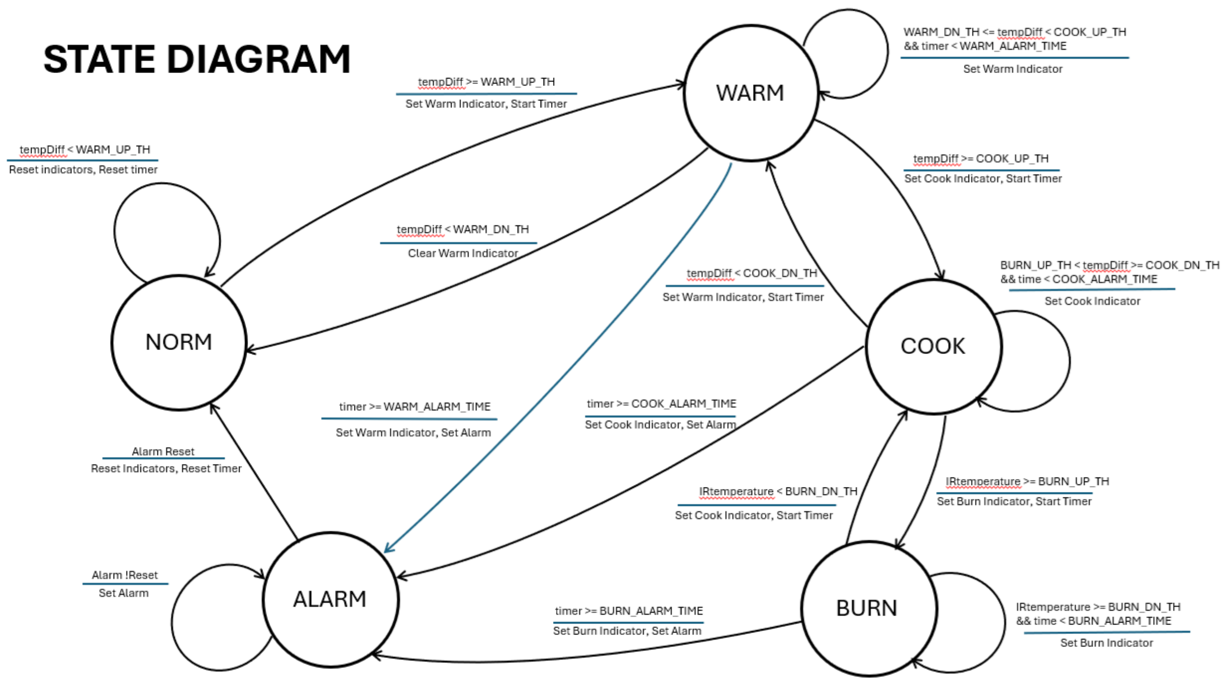

Figure 2. State Diagram.

Referring to figure 2: The system begins in the NORMal state. This state indicates that no burners on turned on. When no burners are turned on, the IR temperature reading is at or below the sensor's ambient temperature reading.

When the IR temperature minus the ambient temperature exceeds a WARM_UP_THreshold, the WARMing state is entered and a timer is initiated. Sensor readings continue as time in state accumulates. If the IR temperature minus the ambient temperature falls below the WARM_DN_THreshold, the timer resets and the system returns to the NORMal state. Conversely, if the IR temperature minus the ambient temperature exceeds a COOK_UP_THreshold, the timer resets and the COOKing state is entered. Finally, if the system stays in the WARMing state longer than a WARM_ALARM_TIME value, the ALARM state is entered and an alarm is generated.

The COOKing state operates similar to the WARMing state, but with its own sets of thresholds for transitioning to either the BURNing state or back to the WARMing state. It also has its own timeout for generating an alarm.

The BURNing state operates similar to the WARMing state but is entered and exited based upon IR temperature only. The threshold for entering the BURNing state is an IR temperature reading that is abnormally high and only occurs when a burner is uncovered on a high setting or else when something is cooking on a high burner setting and has "boiled down" to where it no longer absorbs heat. The BURNing state is abnormal and alarms after a short timeout period.

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The state transition thresholds and timeouts were empirically determined by recording both “naked burner” data on each burner at various heat settings as well as capturing some actual cooking scenarios. Each stovetop, each sensor placement, and each cooking style will likely require different, individualized thresholds and timeouts. For this reason, these constants are stored in a separate “included” file in the program:

> https://github.com/TeamPracticalProjects/Stovetop_Monitor/blob/main/Software/StovetopMonitor/src/ElectricStovetopConstants.h

The microcontroller firmware prints each set of sensor readings to the Serial port on the Photon 2 microcontroller. A computer running a serial monitor program can be used to record various cooking scenarios that can subsequently be analyzed to determine the best thresholds and timeouts for any specific installation.

Note that even in the best case, this non-invasive measurement algorithm is imperfect. We have chosen to err on the side of occasional false positive readings on the basis that false positives are an annoyance, whereas a missed alarm can be dangerous.

The Stovetop Monitor electronics has a green LED, a yellow LED and a red backlit pushbutton that are visible to the user. It also contains a piezo buzzer. The current state of the monitor is indicated as follows:

- GREEN LED on solid: this is the NORMal state. Burners should all be off.
- YELLOW LED on solid: this is the WARMing state.
- YELLOW LED flashing: this is the COOKing state.
- RED LED (pushbutton backlight) on solid: this is the BURNing state.
- RED LED (pushbutton backlight) flashing and Buzzer beeping: this is the ALARM state. The ALARM state can only be exited by either pressing the red push button switch or by tapping the ALARM RESET button on the App.

The Android App can be used to remotely read out the status of the Stovetop Monitor. The App displays:

- The Current State of the Stovetop Monitor.
- The accumulated time in the current state (minutes).
- The latest IR Temperature reading (degrees F).
- The latest ambient temperature reading (degrees F).

The App also contains a button that can be tapped to remotely reset any alarm that is in process.

See the “Stovetop Monitor User Manual” for further details.

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